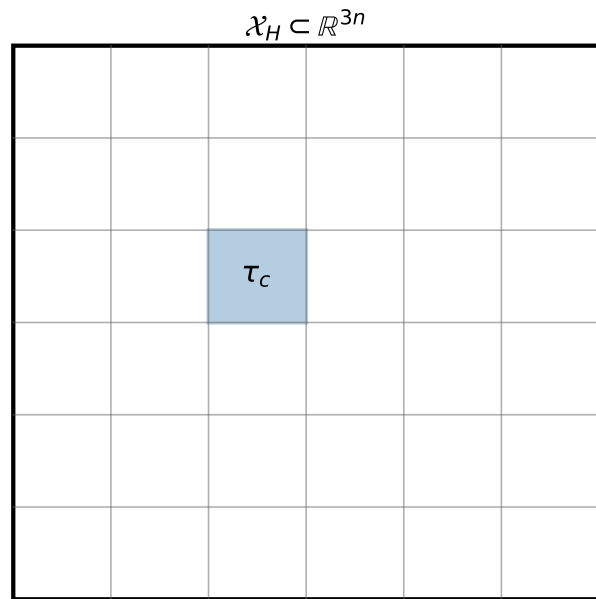


Count estimator of k -fold intersections on the reserve-simplex product

Step 1

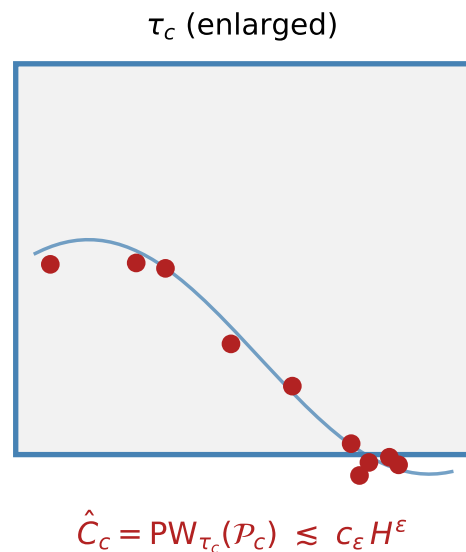
Dyadic partition of reserve-simplex product



cell width $\sim R^{-1/2}$

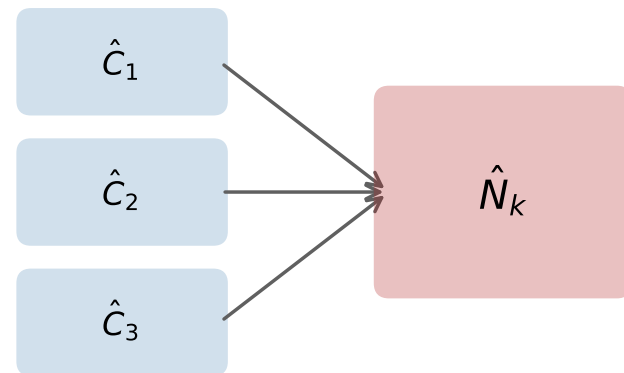
Step 2

Cell-wise Pila--Wilkie count on τ_c



Step 3

ℓ^p composition via counting inequality



$$\hat{N}_k = \left(\sum_c \|\hat{\sigma}_c\|_{L^p}^p \right)^{1/p} \cdot D(R, p, \varepsilon)$$

$$D = O(R^\varepsilon)$$